

MUSTAFA YALÇIN HARATOĞLU

Frontend Developer / Civil Engineer

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Summary

A disciplined engineer who successfully transitioned into frontend development, leveraging a solid foundation in Civil Engineering (100% English) and production management to solve complex architectural challenges in software. Highly adept at translating heavy engineering domains into high-performance web applications by applying core mathematical, spatial, and analytical reasoning. Orchestrated and developed the end-to-end frontend architecture for large-scale corporate Energy Management Systems (EMS/SCADA) and renewable energy simulation platforms (B2B SaaS) at Ratio Energy. Possesses robust, proven expertise within the React 18/19, TypeScript, and Vite ecosystems, specializing in Web Worker-driven real-time data streaming, interactive diagram editors (SynCFusion, React Flow), 3D visualization (Babylon.js), and high-performance analytical dashboard engines.

Professional Experience

03/2024 – Present
Ankara - Remote

Ratio Energy, Frontend Developer

- Architected and engineered the end-to-end web and mobile frontend infrastructure for RatioEMS and RatioSIM products from scratch.
- Real-Time WebSocket Infrastructure: Offloaded WebSocket connections to Web Workers to prevent UI thread blocking; implemented exponential backoff, health-checks, and token-refresh mechanisms to ensure uninterrupted real-time data streaming.
- SCADA Single-Line Diagram Editor: Developed an electrical schematic editor using SynCFusion Diagram, featuring dynamic node/connector property panels, custom signal rules (color/visibility effects), and live data integration.
- General View Editor (Isometric Asset Editor): Designed an advanced asset layout editor on an isometric grid incorporating drag-and-drop asset placement, collision/occupancy detection, auto-routing, and a Babylon.js-driven 3D (GLTF) visualization layer.
- Customizable Dashboard Engine: Built a high-performance dashboard engine leveraging spatial hashing for collision resolution, offering modular drag-and-drop widget placement and a zoom/pan canvas.
- RatioSIM SaaS Development: Engineered a feature-based UI/UX architecture for a 400+ file B2B SaaS platform simulating solar, wind, and battery energy storage (BESS) models; implemented interactive financial sheets (P&L, Cash Flow) and complex energy analytics charts.
- Reporting & Documentation: Programmed a multi-step dynamic PDF report wizard using @react-pdf/renderer. Authored and documented over 30 reusable UI components via Storybook 8 (~60 stories) to establish a comprehensive company design system.
- Test Automation & CI: Authored robust E2E test scenarios for critical user paths (CRUD operations, Authentication) using Playwright and Cypress, integrating them into GitHub Actions CI pipelines.
- Mobile Application: Developed the React Native mobile application for the RatioEMS platform and coordinated deployment workflows across iOS and Android app stores.

01/2020 – 11/2023 Gaziantep	Haratoğlu Pen, <i>Production Manager</i> - Managed end-to-end factory operations and manufacturing processes for a PVC profile extrusion and ready-made door/window assembly facility. - Directed annual production planning, raw material supply chain operations, cost accounting, budget controls, and established strict quality assurance frameworks.
01/2019 – 01/2020 Gaziantep	Griffon Design, <i>Co-Founder & Agency Manager</i> Spearheaded 3D architectural rendering, animation production, and professional commercial/drone filming processes for corporate clients; sharpened visual acuity and the ability to manage complex visual layouts.
06/2017 – 01/2019 Gaziantep	Karen Construction, <i>Site Chief (Civil Engineer)</i> Coordinated the on-site execution of superstructure projects in compliance with technical specifications, budgets, and project timelines; managed subcontractor teams, quantity surveying, progress billing, and occupational safety.

Education

2012 – 2017	B.S., Civil Engineering (100% English), <i>Gaziantep University</i>
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Technical Skills

Frontend Technologies React 19, React 18, TypeScript, JavaScript, Vite, React Router v7, React Native, Zustand, SWR, Axios, i18next	Visualization & Real-Time WebSocket, Web Workers, Syncfusion Diagrams, Babylon.js (3D/GLTF), React Flow, Recharts, react-zoom-pan-pinch, Cesium
UI Frameworks Ant Design 5, styled-components	Testing & Quality Standards Playwright (E2E), Cypress, Storybook 8, ESLint, Prettier, Husky, Commitlint, TypeScript Strict Mode
Tools & Workflows pnpm, Git, GitHub Actions (CI), semantic-release	Domain Knowledge EMS (Energy Management Systems), SCADA, Energy Signals, Simulation Analytics, B2B SaaS, Energy Trading (Bidding/Forecast)
Engineering & Core Competencies Analytical Problem Solving, 3D Spatial & Geometric Reasoning, Cross-functional Team Coordination, Technical Domain Adaptability, Budget & Resource Optimization	

Languages

Turkish Native	English Full Professional Proficiency (Advanced - 100% English Higher Education & Professional Software Engineering Experience)
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Certifications & Training

11/2023 – 02/2024	Full-Stack Frontend Developer Bootcamp, <i>Archis Academy</i> Completed an intensive 6-month curriculum in 4 months with outstanding success and professional certification
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References

Emin Batur Dizdar, *Founding Partner*, Ratio Energy

0538 363 42 20

Thoufeeque (Tofi) Saheer, *CEO & Founder*, Archi's Academy

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